

ARIMA Models: Forecasting Future Movements

ARIMA (AutoRegressive Integrated Moving Average) models are among the most widely used methods for time series forecasting. They combine three components:

- **AR (AutoRegressive):** Uses the relationship between an observation and some number of lagged observations
- **I (Integrated):** Applies differencing to make the time series stationary
- **MA (Moving Average):** Uses the dependency between an observation and a residual error from a moving average model

Let's fit an ARIMA model to Apple's return series and generate a forecast:

```
1. from statsmodels.tsa.arima.model import ARIMA
2. import warnings
3. warnings.filterwarnings('ignore')
4.
5. # Prepare returns data
6. returns = apple['Returns'].dropna()
7.
8. # Fit ARIMA model (p=1, d=0, q=1)
9. model = ARIMA(returns, order=(1, 0, 1))
10. model_fit = model.fit()
11.
12. # Summary of the model
13. print(model_fit.summary())
14.
15. # Forecast next 5 days
16. forecast = model_fit.forecast(steps=5)
17. print("\nForecasted Returns for Next 5 Days:")
18. print(forecast)
19.
```